

Project Initialization and Planning Phase

Date	04 july 2026
Team ID	xxxxxxx
Project Name	Human Development Index(HDI) Predictor
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

The Challenge: Data Inaccessibility

The United Nations Development Programme (UNDP) publishes comprehensive Human Development Reports annually. However, understanding how specific regional factors (like boosting expected schooling by 2 years or increasing GNI by \$5,000) directly impact a country's overall HDI score requires complex, manual geometric mean calculations based on shifting minimum and maximum global goalposts. Furthermore, the raw datasets (often spanning from 1990 to 2022) contain severe data skewness and missing values, making it difficult for non-technical individuals to extract predictive insights without extensive data engineering knowledge.

The Proposed Solution: Algorithmic Approximation

This project proposes the development of a fully automated **Human Development Index Predictor**. By leveraging historical panel data spanning over three decades, we will train a highly accurate Machine Learning regression model capable of instantly approximating the mathematical HDI calculation based on 4 core user inputs. The model will be serialized and deployed via a modern Flask web application, featuring a "Google-grade" Bento Box UI design to ensure maximum accessibility, responsiveness, and educational value.

Project Objectives & KPIs

- **Data Pipeline:** Construct a robust data preprocessing pipeline that dynamically handles missing values and normalizes severe exponential skewness in financial data.
- **Model Accuracy:** Train, evaluate, and optimize a predictive model achieving a minimum R^2 score of > 0.95 and an RMSE of < 0.05 .
- **Deployment:** Deploy a lightweight, production-ready WSGI web server that processes user inputs and visualizes the results dynamically with sub-100ms latency.

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Example:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A data analyst working on HDI prediction.	Predict the Human Development Index accurately.	The dataset contains many variables that affect the prediction.	HDI depends on multiple indicators like education, healthcare, and income, making prediction challenging.	Motivated to build a reliable model that supports better development planning.